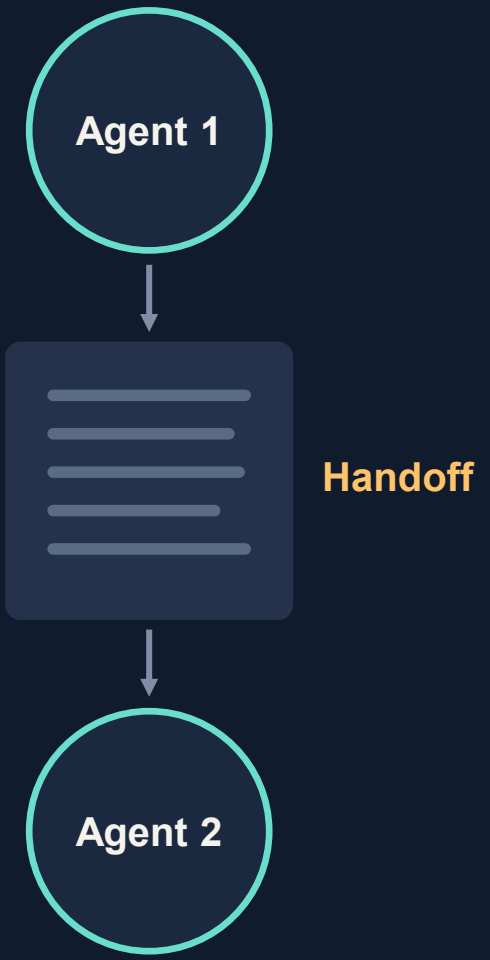
CS3227 AGENTIC SOFTWARE ENGINEERING · WEEKS 8 AND 9

Free-form summaries vs structured checklists

Which handoff format loses less information between AI agents?

An experiment on a real MP3 task



WHY THIS EXPERIMENT

Our MP3 agents pass work along in handoffs

In a multi-agent workflow (Week 8), each agent ends its task by handing off to the next agent. The next agent starts fresh, so the handoff is all it knows.

OUR MP3 WORKFLOW, FOR EACH FEATURE



Each handoff uses a fixed checklist. If it drops something important, every later agent works without it.

GOAL

Find practical ways to keep context when agents hand off work.

This experiment tests one choice: how the handoff is written.

WHY IT IS HARD

A handoff squeezes a long history into a short note

Week 9 calls this context compaction, and warns that compaction is lossy: whatever the handoff leaves out, the next agent never sees.



Message history
every message, tool call and tool result
3,449 to 3,551 words in this experiment

Handoff
200-word limit

Starts fresh
with only the handoff

THE QUESTION

Which handoff format keeps more information?

I compared two ways of writing the same handoff.

FREE-FORM SUMMARY

Connected prose, with no headings or bullet points



STRUCTURED CHECKLIST

Fixed fields: the same ones our MP3 handoffs use

- **Agent and Status**
- **Inputs**
- **Changed**
- **Checks**
- **Decisions needed**
- **Assumptions**
- **Untrusted content seen**

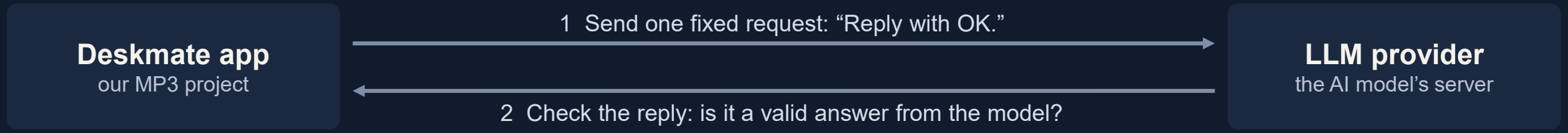
Both formats get the same instructions, the same message history and the same 200-word limit.

Then I measured **how much information the next agent correctly recovers, misses or gets wrong.**

THE CODING TASK

The task: fix five bugs in Deskmate’s LLM health check

First, a coding agent did a real MP3 task. Deskmate’s health check tests whether our app can reach its LLM provider, and its starting code had five bugs.



FIVE BUGS IN THE STARTING CODE

1

Timeout

No overall 10-second limit on the whole check

2

Proxy

May send the request via the machine’s proxy server

3

Extra header

Sends a User-Agent header it should leave out

4

Bad access tokens

Some crash the check instead of being refused

5

Cut-off replies

Crash the check instead of counting as a failure

The fix also had to keep every other documented rule, such as which replies count as a success.

THE SETUP

Three agents pass the work along

The coding agent's message history goes to two handoff agents, one per format. A fresh receiving agent then answers six questions using only one handoff.



5 coding runs × 4 handoff models × 2 formats = 40 handoffs, each quizzed once

The 4 handoff models: GPT-5.6 Sol, GPT-5.6 Luna, Claude Opus 5.5 and Claude Sonnet 5.5

SCORING

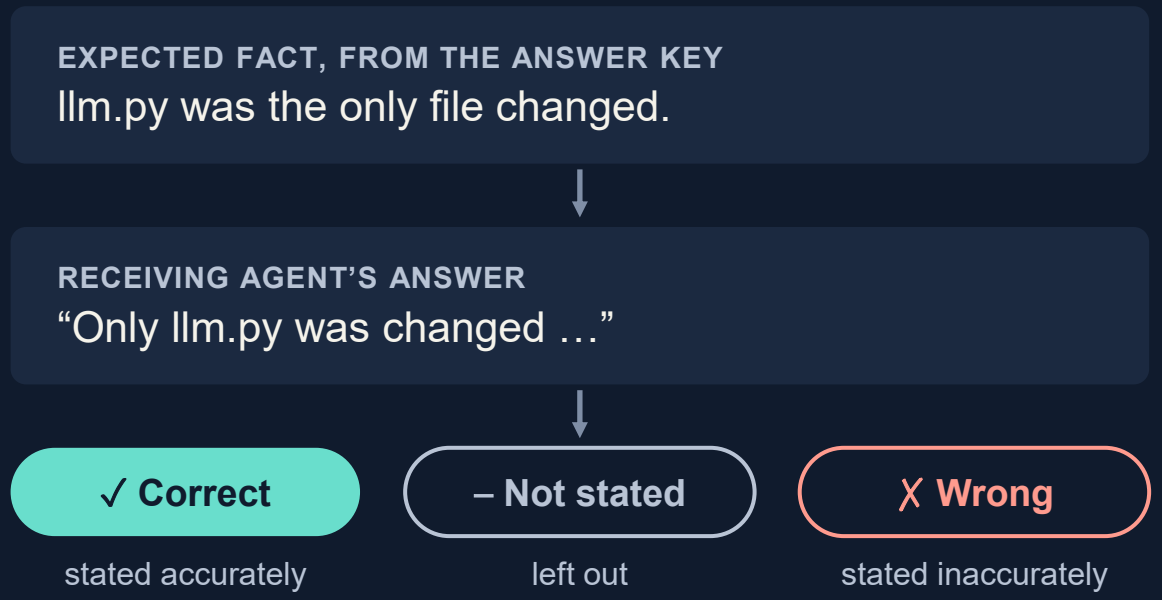
Scoring: how many expected facts survived?

For each coding run, I wrote an answer key before its handoffs existed. It splits the six questions into expected facts, taken only from that run's message history.

THE SIX QUESTIONS

1. Which files were changed?
2. Which checks were run, and what happened?
3. What did the agent leave unverified?
4. How and why was the timeout fixed?
5. Which provider responses count as a success?
6. What was recorded about proxies and headers?

HOW ONE FACT IS SCORED



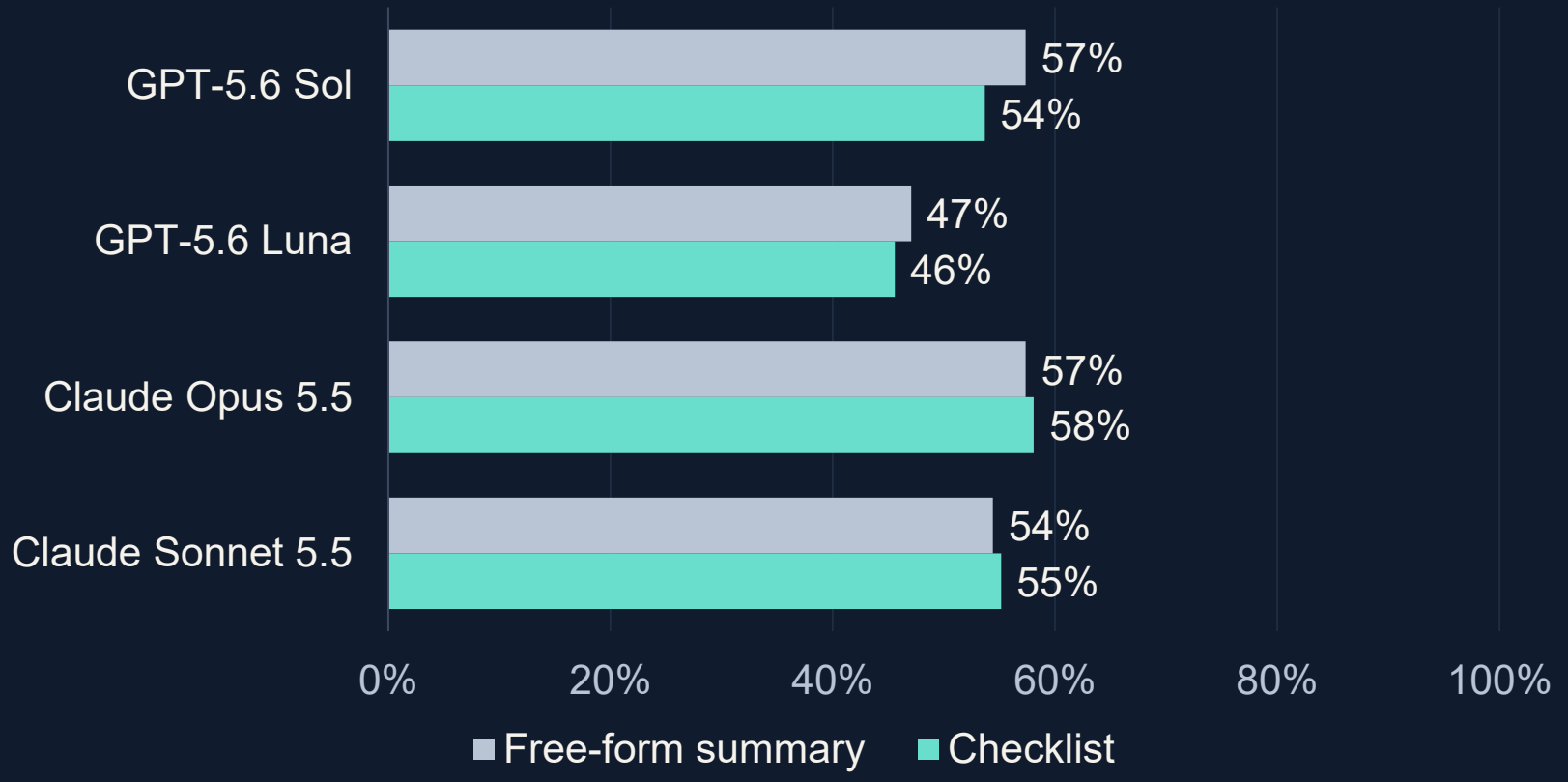
Score = correct facts ÷ expected facts

136 expected facts for each handoff model and format

RESULT 1

GPT leaned free-form, Claude leaned checklist

No format won overall. GPT models did slightly better with free-form summaries, and Claude models with checklists. Every gap was 5 facts or fewer, out of 136.



GPT: summary ahead
by 5 facts (Sol) and 2 facts (Luna)

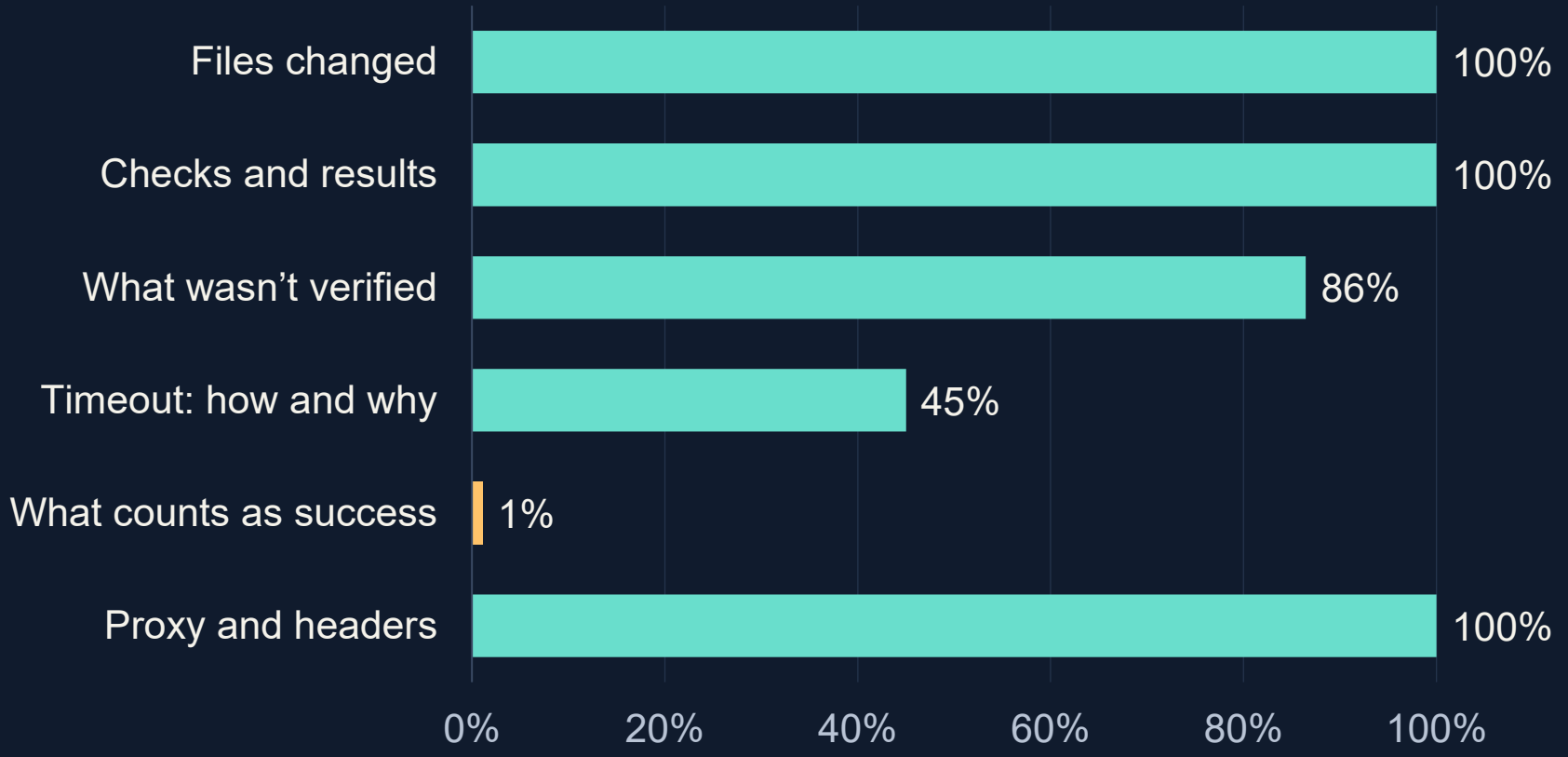
Claude: checklist ahead
by just 1 fact each (Opus and Sonnet)

Treat these gaps with care
9 of the 10 Claude checklists broke the 200-word limit, and two extra runs flipped Luna’s lead.

RESULT 2

The real gap was in which facts survived

Facts about what the agent did nearly always survived. The rules for what counts as a successful response almost never did.



4 of 360

facts about what counts as success survived, across all 40 handoffs

Why?

The handoffs described what the agent changed. They almost never listed the rules the check had to meet.

AN EXAMPLE

The checklist kept the test results but lost the rules

This real checklist reported that all 30 tests passed. But it never said which responses count as a success, so the receiving agent answered “not stated”.

THE HANDOFF

GPT-5.6 Sol checklist, coding run 4

Checks: Offline standalone suite passed all 30 tests in 0.030s. Covered request shape/headers, tokens, response validation, 65,536-byte limit, …

✓ **Test results kept**

✗ **Nothing on which responses count as a success**

THE RECEIVING AGENT

Asked

“Which provider responses count as a successful probe?”

Answered

“not stated. The handoff mentions only response validation and shape validation, without saying which responses count as success.”

THE ANSWER KEY

9 expected facts for this question

- ✗ Status code is 200
- ✗ Body is valid UTF-8
- ✗ Body is valid JSON
- ✗ Has a non-empty “choices” list
- ✗ First choice has a text message
- ✗ Message is not blank
- ✗ Message need not say “OK”
- ✗ Content-Type header is ignored
- ✓ **Body is at most 65,536 bytes**

1 of 9 recovered

LIMITATIONS

What this experiment can’t tell us

These results describe this one setup. They don’t show that either format is better in general.

1 — **One coding task**

And a quiz, not real follow-up work.

1× — **One handoff per condition**

Each was written once, with no repeats.

≈ — **The formats ended up alike**

Most summaries still labelled each part inline.

12 — **Some went over 200 words**

12 of 40, including 9 of the 10 Claude checklists.

+2 — **Two runs added late**

Runs 4 and 5 came after I saw runs 1 to 3.

AI — **AI grading, not blind**

The grader could see each handoff’s format.

TAKEAWAY

Decide what goes into a handoff, not just how it looks

The format made only small differences. What mattered more was whether the handoff had the right facts.

RECALL FIRST · WEEK 9

Lost information can’t be recovered. Extra information can be trimmed later.

Our handoffs kept what the agent did, but lost the rules the work had to meet.

IDEAS TO TEST NEXT, FOR OUR MP3 WORKFLOW

1 **Add a goal and success-criteria field**

Structured note taking (Week 9): the lecture’s note template starts with the Goal; our checklist has no such field.

2 **Point to the requirements**

Just-in-time retrieval (Week 9): link the contract, so the next agent can look the rules up itself.

Neither idea was tested in this experiment.

BACKUP

Full results by model and format

Correct facts out of 136 expected facts, across all five coding runs. No expected fact was scored wrong; the rest were not stated.

HANDOFF MODEL	FORMAT	CORRECT FACTS	SHARE	OVER 200 WORDS
GPT-5.6 Sol	Free-form summary	**78 of 136**	57%	0 of 5
GPT-5.6 Sol	Checklist	**73 of 136**	54%	0 of 5
GPT-5.6 Luna	Free-form summary	**64 of 136**	47%	0 of 5
GPT-5.6 Luna	Checklist	**62 of 136**	46%	0 of 5
Claude Opus 5.5	Free-form summary	**78 of 136**	57%	3 of 5
Claude Opus 5.5	Checklist	**79 of 136**	58%	5 of 5
Claude Sonnet 5.5	Free-form summary	**74 of 136**	54%	0 of 5
Claude Sonnet 5.5	Checklist	**75 of 136**	55%	4 of 5

BACKUP

How the test was kept fair

Full method, exclusions and audit details are in report.html inside the results ZIP.

Same start for every coding run

A fresh copy of the same starting code, contract and 30 offline tests. Edit llm.py only; no network. All five repairs passed all 30 tests.

Separate agents, no peeking

Fresh sessions. Handoff and receiving agents had no tools. No agent saw the other handoff or the answer key.

Fixed before the handoffs

Questions, prompts and each run’s answer key were frozen before that run’s handoffs were written. Runs 4 and 5 reused the same questions and prompts.

Changes from the original plan

The plan named GPT-6.1 Sol for the coding and receiving agents; the runs used Claude Opus 5.5. Five coding runs instead of three; two were added after the first results.